

# Statistical and Computational Models

## Lecture 4: Assessing models

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## A quick recap

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How do we characterise that trade off?

# Roadmap

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The one sample problem

The validation split

Cross-validation

The test set

## The missing test curve

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The way out is one we already know: **hold out**. Fit on some rows, score on the others.

## Refining the training MSE

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Better approaches to estimating the EPE, using  $\mathcal{D}$  alone:

1. Validation set approach.
2. Cross-validation.
3. Test set approach (ideal).
4. 'In-sample' adjustment – next lecture.

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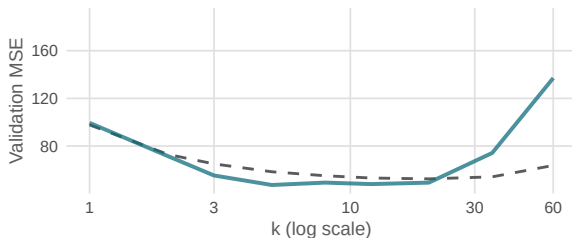
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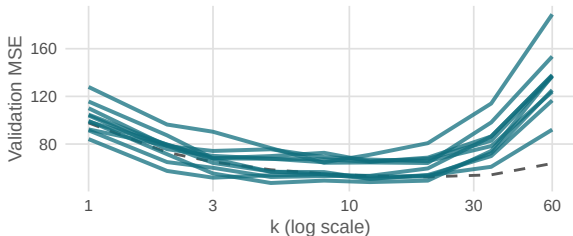
It can also run conservative, overestimating the error. **Why?**

## The validation split in practice



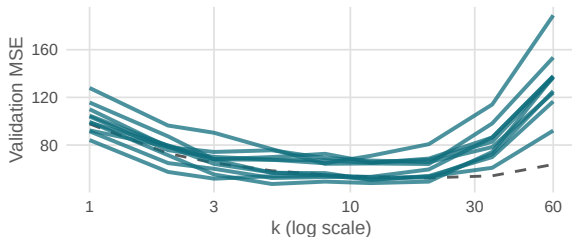
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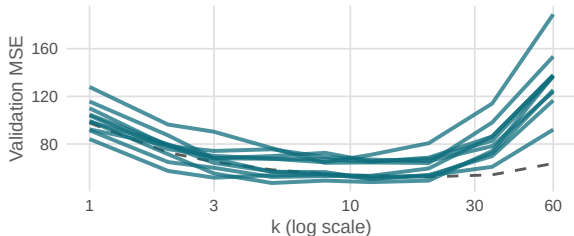
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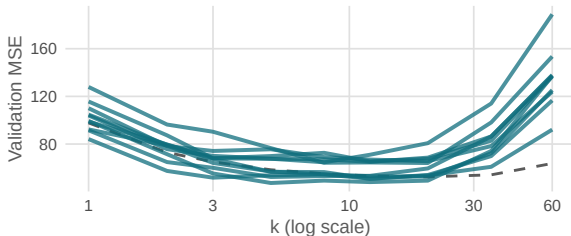
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They also pick poorly: 7 of 10 put their minimum below  $k = 20$ , which we know to be the optimal  $k$ .

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## Method 2: Leave-one-out cross-validation

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### Method 2a: leave-one-out cross-validation

For each  $i = 1, \dots, n$ :

1. Move observation  $i$  to the validation set.
2. Fit on the remaining  $n - 1$  observations.
3. Score observation  $i$ .

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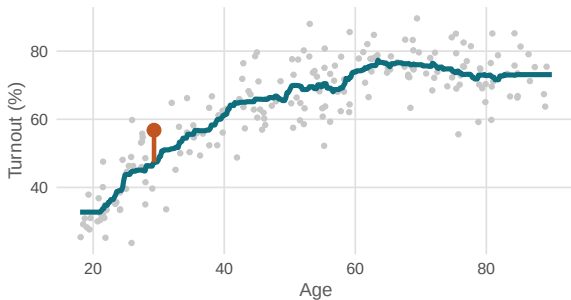
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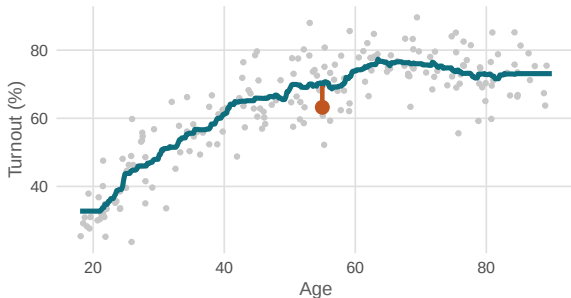
Each fit now sees  $n - 1$  observations, and the power of chance is dramatically reduced. This solves both of our previous problems. **Why?**

## LOOCV in practice



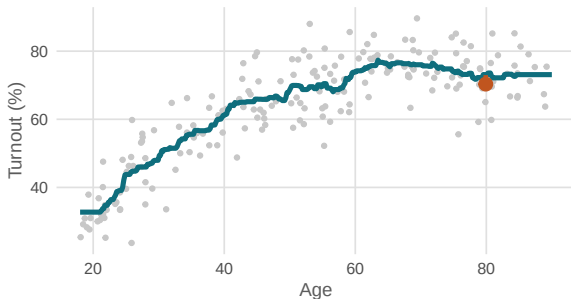
Hold out the orange observation; fit on the other 199.

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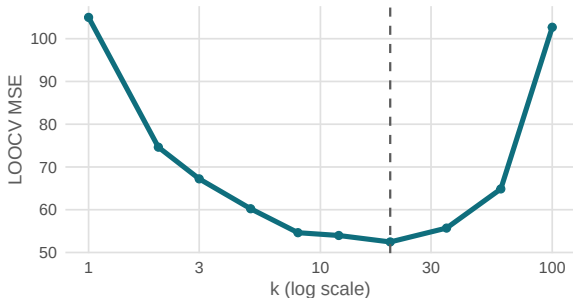
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Average the 200 errors at each  $k$  to generate the LOOCV curve. Its minimum sits at  $k = 20$  (nice).

## Method 2: $K$ -fold cross-validation

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### Method 2b: $K$ -fold cross-validation

1. Divide  $\mathcal{D}$  into  $K$  **fold**s:  $\mathcal{I}_1, \dots, \mathcal{I}_K$ .
2. For each fold  $j$ : fit on the other  $K - 1$  folds, score fold  $j$ , call its loss  $\text{MSE}_j$ .

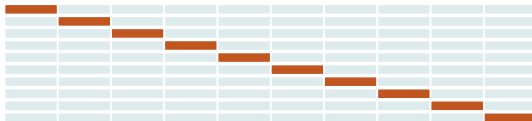
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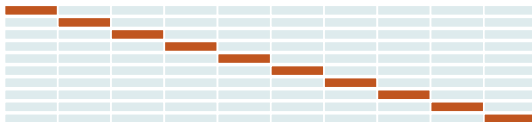
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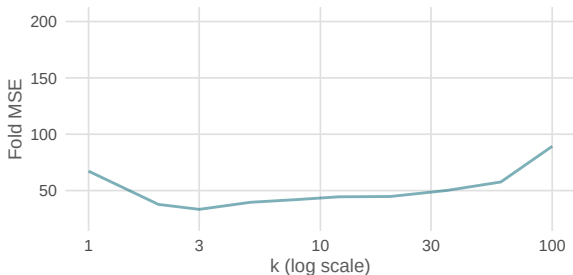
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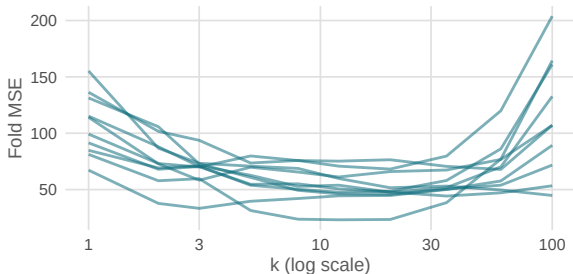
Note: capital  $K$  counts folds; lower-case  $k$  still belongs to the neighbours. LOOCV is  $K = n$ ; the validation split was  $K = 2$  with no rotation. So what should  $K$  be?

## K-fold CV in practice



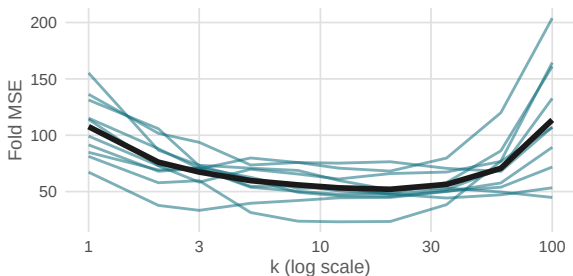
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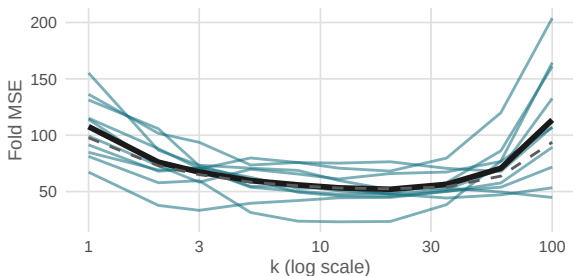
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Dashed is the DGP  $\mathbb{E}[EPE]$  – the CV still runs a little high due to the sample size sacrifice.

## K-fold CV machinery

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```
set.seed(6039)
K <- 10
fold <- sample(rep(1:K, length.out = n))
mse <- matrix(0, K, length(ks4))
for (j in 1:K) {
  tr <- train[fold != j, ]           # nine folds
  va <- train[fold == j, ]           # the tenth
  mse[j, ] <- colMeans((va$y - knn_all(tr, va$x)[, ks4])^2)
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Whatever our chosen loss is what we calculate across units. If the loss is wrong for the problem, CV optimises the wrong thing.

## Choosing $K$

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### Bias:

- With few folds each fit sees far fewer than  $n$  observations, so  $CV_{(K)}$  runs high.
- How much higher depends on whether the estimator is still improving between  $n(K - 1)/K$  observations and  $n$ . Here, with 10 folds, hardly at all.

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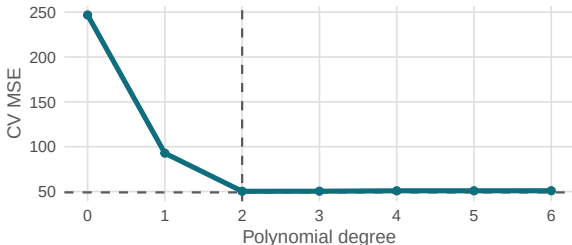
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$CV_{(n)}$  – LOOCV – carries the least bias and the most variance;  $CV_{(2)}$  – the validation split, rotated once – does the opposite.

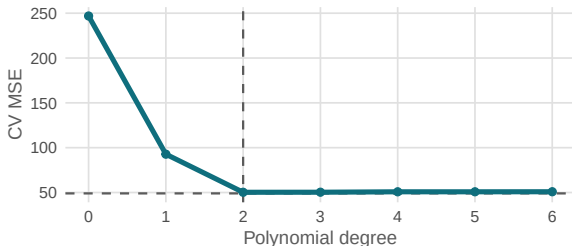
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CV picks  $d = 2$ ; anything more flexible than  $d = 2$  sits at the irreducible error floor,  $\sigma^2 = 49$ .

## The CV standard error

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In our case, if we redraw the folds 50 times,  $k = 20$  wins 49 times. The minimum stays between 51.4 and 55.9.

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Temptation:

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Is this a correct application of cross-validation?

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Question: why do the last two disagree? The fold must redo **every** data-dependent step: screening, standardising, imputing, choosing  $k$ . Failing to do this caused “leakage”.

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## Method 3: Test set

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1. Randomly divide  $\mathcal{D}$  into three sets: training, validation, and test.
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This requires a lot of discipline. You need lots of data, you sacrifice data, and you lose researcher degrees of freedom.

## Two tasks

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**Model assessment:** how good is our chosen model?

# References

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- James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. – **5.1.** statlearning.com.
- Lundberg, I., Johnson, R. & Stewart, B. M. (2021) "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." *American Sociological Review* 86(3): 532–565.
- Hastie, T., Tibshirani, R. & Friedman, J. (2009) *The Elements of Statistical Learning*, 2nd ed. – 7.10, 7.12.